Linguistic Misinformation Markers

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Abstract

We investigated linguistic markers, including adjectives, capitalization, exclamation, hedging, and profanity, as signals for automated classification of misinformation and opinion in short social media and news items. We extended the Hugging Face Twitter Misinformation Dataset with linguistic feature annotations and opinion labels across 1,000 examples. Using this corpus, we trained and evaluated models including logistic regression and convolutional neural network (CNN) architectures, and explored transfer learning, multi-task learning with POS features, bootstrapping, active learning, few-shot learning, and ensemble methods. Our best misinformation classifier, a Lean Motivated Ensemble, achieved a Macro F1 of 0.918, and our best opinion classifier achieved a Macro F1 of 0.761.

1 Introduction

The rise of social media and non-traditional news outlets has fundamentally changed the information disseminated to the public. While internet platforms have democratized publishing, they have also made it significantly harder for readers to distinguish factual reporting from misinformation and opinion. Unlike traditional journalism, which operates with editorial standards and fact-checking processes, social media posts and online news items may be published with little or no verification. The reader's challenge is compounded by the fact that digital text items rarely fit into simple categories of "true" or "false." An editorial may contain accurate facts alongside personal opinion. A social media post may blend an emotional narrative with a misleading claim or true facts.

We believe that linguistic patterns offer a signal for this classification problem. Writing that aims to mislead or persuade tends to differ stylistically from writing that aims to inform. Fact-based news and false information or opinionated texts likely have measurable differences in linguistic features such adjectives, emotional language, capitalization, vague sourcing, and clickbait phrasing, depending on whether the text aims to disseminate factual information, spread misinformation, or express emotions and opinions. In addition to developing models to classify writing containing misinformation vs fact, we also developed models to classify writing that contains opinion vs fact.

These linguistic differences may be subtle, but they are potentially measurable. NLP and machine learning methods are well-suited to detecting such patterns at scale, offering a path toward automated tools that could assist readers, platforms, and researchers in assessing the credibility of online content. In addition, a secondary objective of this project is to contribute an enhanced dataset to Hugging Face, so it can be used by other researchers studying how language signals credibility.

2 Data

As a foundational corpus, we used the Hugging Face Twitter Misinformation Dataset (Minassian). This dataset includes text and a misinformation label as columns. We altered the corpus by annotating 1,000 examples to include labels to create the following features:

1. `adjectives`: Adjectives with suffixes -ful, -less, -ment, -ness, -ing, or -ible.
2. `all_caps`: Capitalised words that are not acronyms.
3. `exclamation_marks`: Any number of exclamation marks.
4. `hedging`: Words found in the hedging lexicon (hedging lrec).
5. `unk`: Profanity.

In addition, we removed duplicate text entries. Texts were deemed to be duplicates when only the URL was modified, likely due to tokens added when choosing to share to Twitter/X from a third party website. To ensure the hand-annotated dataset was manageable, we limited text length to 560 characters, and truncated text items when necessary. Since each example in the corpus was already labeled with 0 (factual) or 1 (misinformation), we added a column for opinion, with a value of 1 for "opinion" and 0 for "not opinion."

Given the small size of the dataset, we decided on a 60-20-20 train-validation-test split. This was conducted using stratification, to ensure proportions in each split reflect the main dataset in terms of distribution of misinformation (35% misinformation, 65% factual) and opinion (41% opinion, 59% not opinion). We also included each main annotator as a stratification factor, being annotators that individually represented at least 20% of all annotations (Jennifer, Nicole, Shiao-li, and Rachelle). The remaining three annotators (AI, Majority Vote, and Jasmine) were not included in the stratification to avoid imbalance. To ensure reproducibility, the dataset splits are added as a column to the main dataset and the script sets a random seed to 344.

For bootstrapping, we used `df.sample` to select 250 additional rows from the cleaned dataset, still limiting to 560-character text limits and no exact match/duplicates in text before the URL.

3 Methodology

3.1 Baseline Models

3.1.1 Traditional Method: LogReg

For our traditional model, we used logistic regression. This model, LogReg, is appropriate for the text classification task due to its scalability for sparse features and training speed.

To evaluate the model's ability to identify opinions, text and numeric features were used. TF-IDF vectorization was applied over unigrams and bigrams for text data assessment. Additionally, our annotated linguistic features were scaled using StandardScaler before combining them with the TF-IDF matrix. The TF-IDF captures vocabulary and content patterns, while the annotated linguistic features identify writing style markers. Therefore, the inclusion of both feature types was necessary to adequately measure the differences between factual information, misinformation, and personal opinions. To account for imbalances between these classes, the model was trained with balanced class weights.

For hyperparameter 'C', we used GridSearchCV to conduct 5-fold cross-validation on the training set. We used the macro F1 score to select the appropriate C value to account for class imbalances.

3.1.2 Neural Method: TextCNN

For the baseline neural model, referred to as TextCNN, a Convolutional Neural Network (CNN) was used in conjunction with NLTK's tweetTokenizer and FastText frozen embeddings. For the baseline, the text was the only part of the corpus used for training. Since the example texts in the corpus are of short length, opinion signals may exist in 3-gram, 4-gram, or 5-gram sequences, rather than in longer sequences. CNN was chosen because it relies on local markers by using a shifting window of n-grams to find patterns in text. In addition, given the low resource constraints of the project, CNN is robust against overfitting. Hyperparameters were chosen according to recommendations in "Convolutional Neural Networks for Sentence Classification," (Kim, 2014) which recommends filter windows (h) of 3, 4, 5 with 100 feature maps each, dropout rate (p) of 0.5, and l2 constraint (s) of 3.

Both the tokenizer and word embeddings were chosen because of their specialization in transforming internet text. NLTK's tweetTokenizer was used because it was developed to handle text boundaries that are unique to Twitter, including hashtags and emojis. Finally, FastText embeddings were selected because they are robust against typos and slang by using subword n-grams. The FastText embeddings were frozen to prevent additional overfitting of the training data, since the training set only included 600 examples.

4 Experimentation

4.1 Ensembling

4.1.1 Soft Vote Ensemble

We first implemented a simple ensemble using the top LogReg model and the top CNN model. Soft voting was implemented with each model's predicted probabilities averaged equally, then applied a decision threshold of 0.5 for the final averaged prediction. This preserved model confidence in its class prediction rather than relying on hard-voting.

4.1.2 Motivated Ensemble

This ensemble method was selected to address class imbalances and differences in model performance and used for the top LogReg and top CNN models. The predicted probabilities were weighted by the Macro F1 score and the decision thresholds were tested with 0.05 steps from 0.3 to 0.7. The threshold remained 0.50 for opinion classification and the misinformation classification used a 0.40 threshold.

4.2 Transfer Learning

4.2.1 LogReg (+ embeddings)

To augment the LogReg model, we applied the same FastText frozen embeddings as the TextCNN model. It is worth noting that embeddings typically perform better in non-linear models like neural networks. To balance the consideration of TF-IDF and embeddings to reduce feature imbalance, TF-IDF weighted pooling was applied to give higher weight of embeddings to more important words in corpus.

4.2.2 TextCNN Transfer

Since TextCNN already included transfer learning via FastText frozen embeddings, we implemented an additional transfer learning signal by implementing a model that uses a single convolutional backbone across both misinformation and opinion classification tasks. The backbone is trained on the text of each example to find patterns that are jointly useful for predicting opinion and misinformation, rather than optimizing for either label in isolation. The signal from one task provides secondary supervision to the other.

4.3 Multi-Task Learning: LogReg MTL(cascaded POS), TextCNN MTL(POS), TextCNN MTL(POS + linguistic)

We used Part-of-speech (POS) tagging to enhance LogReg and TextCNN through Multi-Task learning. POS tagging was selected because opinion tends to feature more pronouns, adjectives, and adverbs than factual reporting. Factual reporting relies more on nouns and verbs. Since we are also training models for misinformation, we wanted to investigate whether our misinformation models, which were already robust, would also improve with POS tagging.

Before this experiment, TextCNN had only trained on the text of the corpus. In this experiment, we trained TextCNN with the linguistic marker features and POS. In addition, we trained TextCNN with POS only.

4.4 Bootstrapping

In order to perform bootstrapping, we started with our best models for each task. For opinion, the best model was the baseline TextCNN, and for misinformation, Logistic Regression Transfer (with embeddings). Using the respective trained models, predictions were generated and saved for 250 additional rows as silver data. These predictions were added as training data, used to re-train the respective models and then evaluated on the development set.

4.5 Active Learning

To perform active learning, we began with the same 250 bootstrap examples generated in Section 4.4. Each example was scored for annotation uncertainty using a combined metric: the average of a margin-based uncertainty score (how close the model's predicted probability was to 0.5) and a committee disagreement score (whether the top LogReg and TextCNN models disagreed in their hard predictions) across both tasks. Examples were then ranked by this combined score in descending order, and the ranked list was exported for manual re-annotation. The top-ranked examples were re-annotated by hand using the same guidelines as the original dataset. We then ran an accumulation test at 5%, 10%, 15%, and 20% of the 250 examples. We replaced model-assigned silver labels with human labels for the top-k rows at each threshold, while retaining model labels for the remainder. At each threshold, the best opinion model (TextCNN) and best misinformation model (LogReg + embeddings) were retrained on the augmented training set and evaluated on the development set.

4.6 Few-Shot Learning

To further expand the training corpus, we used Gemini AI to generate synthetic training examples equal to 20% of the original dataset size, bringing the total augmented dataset to approximately 150% of the original. We prompted Gemini AI with the augmented training dataset, the annotation schema and a hedging lexicon, instructing it to balance the four possible combinations of classes in the generated output. We also provided false positive and false negative data samples to address samples our models found challenging to classify. (See Figure A.1 in the Appendix for the Prompt-Response Log). Each synthetic example was constrained to the same 560-character limit used in the original corpus and assigned labels for both misinformation and opinion. The generated examples were not subject to human re-annotation. The synthetic examples were appended to the already augmented training set and used to retrain the best-performing models from each family: TextCNN for opinion classification and LogReg (+ embeddings) for misinformation classification.

5 Results

5.1 Metrics

To evaluate each model's performance, we used Macro F1 as the primary metric, with AUC-ROC as the secondary metric. Since our data has a class imbalance, Macro F1 was selected as the primary metric since it weighs each class equally. We included AUC-ROC scores as a secondary metric to assess the model's ranking of text examples by confidence, independent of the decision threshold of 0.5. Lastly, we reported F-beta scores at beta=1.5 to penalize false negatives with the motivation that incorrectly tagging misinformation or opinion as factual causes greater harm.

5.2 Misinformation Classification Results

In misinformation classification, the two ensemble methods performed best with Motivated Ensemble (Macro F1 = 0.9180) and Soft Vote Ensemble (Macro F1 = 0.9171). The best performance in Macro F1, F1.5, and AUC-ROC by ensembling the highest performing LogReg model and CNN model suggests that the two families of models are capturing somewhat different signals that are stronger when combined. Among individual models, LogReg with embeddings and AL achieved the next highest Macro F1 of 0.9031, with all AL thresholds (5%, 10%, 15%, 20%) yielding identical Macro F1 scores, suggesting that even a small amount of human-annotated silver data meaningfully improved performance. However, further annotation did not provide additional gains on this task. Few-shot learning, transfer learning and multi-task learning with POS and linguistic features offered modest improvements over the baseline LogReg and TextCNN models, though the gains were smaller than those achieved through ensembling. Table 1 summarizes the results for all models.

Table 1: Misinformation Models' Results

Model	Macro F1	F1.5	AUC
Motivated Ensemble	0.9180	0.8966	0.9644
Soft Vote Ensemble	0.9171	0.8881	0.9647
LogReg (+ emb) AL 10%	0.9031	0.8603	0.9493
LogReg (+ emb) AL 5%	0.9031	0.8603	0.9476
LogReg (+ emb) AL 20%	0.9031	0.8603	0.9474
LogReg (+ emb) AL 15%	0.9031	0.8603	0.9471
LogReg (+ emb) Bootstrap	0.8973	0.8553	0.9474
LogReg (+ emb)	0.8889	0.8318	0.9501
LogReg (+ emb) FSL	0.8859	0.8125	0.9693
TextCNN MTL (POS + ling)	0.8845	0.8363	0.9553
LogReg MTL (cascaded POS)	0.8831	0.8269	0.9157
TextCNN	0.8816	0.8174	0.9591
TextCNN MTL (POS)	0.8816	0.8174	0.9587
LogReg	0.8784	0.7975	0.9630
TextCNN Transfer	0.8684	0.7979	0.9608

5.3 Opinion Classification Results

Similarly, in opinion classification, the two ensemble methods were the best performers, with both scoring Macro F1 = 0.7613. Aside from the ensemble models, the CNN models consistently beat the LogReg models in this task. This result suggests that opinion classification benefits from a neural learning approach, likely because opinion signals are distributed across local n-gram patterns that CNN architectures are better suited to capture than the sparse TF-IDF features used by LogReg. Since TextCNN already included frozen FastText embeddings, the baseline already included some transfer learning. Multi-task learning, additional Transfer learning, AL, few-shot learning, and bootstrapping provided mixed results on this task, with all models underperforming, relative to the baseline TextCNN. These results suggest that the silver data introduced noise that offset the benefit of additional training examples. Table 2 includes the summary of all opinion classification results.

Table 2: Opinion Models' Results

Model	Macro F1	F1.5	AUC
Motivated Ensemble	0.7613	0.7462	0.8031
Soft Vote Ensemble	0.7613	0.7462	0.8027
TextCNN	0.7410	0.7229	0.7992
TextCNN AL 10%	0.7395	0.7097	0.7875
TextCNN FSL	0.7395	0.7097	0.7808
TextCNN MTL (POS)	0.7383	0.7454	0.7852
TextCNN MTL (POS + ling)	0.7374	0.7330	0.7788
TextCNN AL 20%	0.7320	0.7241	0.7850
TextCNN Transfer	0.7314	0.7177	0.7827
TextCNN AL 5%	0.7308	0.7112	0.7880
TextCNN Bootstrap	0.7267	0.7152	0.7816
TextCNN AL 15%	0.7225	0.7191	0.7541
LogReg (+ emb)	0.7023	0.6962	0.7698
LogReg MTL (cascaded POS)	0.6963	0.6806	0.7486
LogReg	0.6931	0.6530	0.7514

6 Conclusion

We investigated linguistic markers as signals for automated classification of misinformation and opinion in short social media and news texts. By extending the Hugging Face Twitter Misinformation Dataset with annotations for adjectives, capitalization, exclamation marks, hedging, and profanity, we demonstrated that linguistic features offer a meaningful signal for both tasks, with ensemble methods consistently outperforming individual models. Our misinformation classifiers were more successful than our opinion classifiers, with the best model scoring approximately 0.16 Macro F1 points higher. This performance gap reflects the inherently subjective nature of the opinion label. Furthermore, misinformation correlates more strongly with surface-level stylistic signals, while opinion is subtler and more context-dependent. Digital texts rarely fall into clean categories of true, false, or opinionated.

Future work could explore larger annotated datasets, additional linguistic features, and transformer-based architectures to improve opinion classification, in particular. As social media and non-traditional news outlets continue to blur the line between factual reporting and persuasion, automated tools that can assist readers, platforms, and researchers in assessing content credibility become increasingly important. We hope the enhanced dataset contributed to Hugging Face will support further research in this direction.

7 Limitations

Aside from the time constraints of our project and the limited amount of annotated data, we encountered limitations with respect to cross-platform computing. Due to the disk space required for frozen FastText embeddings, some of the team's computers could not train the models that required the embeddings. Once we discovered the issue, we switched to training the models using Google Colab T4 GPUs. Switching the compute environment caused notable differences in the final metrics of models that directly or indirectly rely on CNN. The differences were caused by the implementation of floating point arithmetic in T4 versus Apple M-series silicon. Since some of our previous reports include statistics that are different than the ones in the final report, we are listing the differences here in Tables 3 and 4.

Table 3: Misinformation task: model performance by hardware

Hardware	Macro-F1	F1.5	AUC
Motivated Ensemble			
Apple Silicon	0.9161	0.8794	0.9676
Colab T4	0.9180	0.8966	0.9644
Soft Vote Ensemble			
Apple Silicon	0.9161	0.8794	0.9676
Colab T4	0.9090	0.8654	0.9644
TextCNN			
Apple Silicon	0.8773	0.8221	0.9528
Colab T4	0.8816	0.8174	0.9591
TextCNN MTL (POS)			
Apple Silicon	0.8758	0.8125	0.9608
Colab T4	0.8816	0.8174	0.9587
TextCNN MTL (POS + linguistic)			
Apple Silicon	0.8627	0.7932	0.9489
Colab T4	0.8845	0.8363	0.9553
TextCNN Transfer			
Apple Silicon	0.8742	0.8027	0.9593
Colab T4	0.8684	0.7979	0.9608

8 Ethical Considerations

8.1 Dataset Bias

Potential bias is inherent in our use of the Twitter Misinformation dataset. In addition to having a demographic skew, Twitter is dominated by English language posts. The linguistic marker techniques we explored were developed for use in conjunction with English language text. Furthermore, the dataset contains misinformation labels that were assigned to text during a specific period of time and with a specific political context, and were crowdsourced. As such, the misinformation labels provided by the foundational dataset cannot represent ground-truth.

8.2 Dataset Privacy

The Hugging Face Twitter Misinformation dataset includes Twitter data that was aggregated with no identification of individuals.

8.3 Annotation and Label Biases

Since our team annotated the linguistic markers and opinion labels, they contain our team's biases. There is cultural variability in defining what words or phrases constitute profanity and other linguistic markers. Furthermore, labeling text as "opinion" or "not opinion" is a subjective task requiring individual judgment on subtle word usage.

Table 4: Opinion task: model performance by hardware

Hardware	Macro-F1	F1.5	AUC
Motivated Ensemble			
Apple Silicon	0.7347	0.7666	0.7989
Colab T4	0.7613	0.7462	0.8031
Soft Vote Ensemble			
Apple Silicon	0.7177	0.7165	0.7985
Colab T4	0.7613	0.7462	0.8027
TextCNN			
Apple Silicon	0.7261	0.7087	0.7930
Colab T4	0.7410	0.7229	0.7992
TextCNN MTL (POS)			
Apple Silicon	0.7362	0.7203	0.7913
Colab T4	0.7383	0.7454	0.7852
TextCNN MTL (POS + linguistic)			
Apple Silicon	0.7172	0.7101	0.7917
Colab T4	0.7374	0.7330	0.7788
TextCNN Transfer			
Apple Silicon	0.6963	0.6806	0.7701
Colab T4	0.7314	0.7177	0.7827

8.4 Limitations of Linguistic Markers

The linguistic markers we explored are proxies. The are stylistic features of text that may correlate to misinformation, opinion, and fact. The linguistic markers themselves are not misinformation, opinion, or fact indicators. A well-written, subtle piece of misinformation or opinion could escape correct classification.

References

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A LLM Prompt-Response Log

Figure A.1: LLM prompt-response log used for training data generation.

Prompt: I'm attaching three documents, please make sure you understand their format and content first:

1. existing training set
2. annotation schema
3. lexicon of hedging words

Output: I have reviewed the three documents you provided. Here is a summary of their format and content:

1. Training Set (training_bs_al.csv) This is a CSV file containing social media posts (likely from Twitter/X) related to natural disasters and other events. It includes the following key columns: *text*: The raw text of the post. *misinformation_label* & *opinion_label*: Binary classifications for the content. Annotation Columns: Columns like *all_caps*, *exclamation_marks*, *hedging*, *adjectives*, and *unk* contain list-formatted data indicating specific spans of text that match the schema (e.g., [['PLEASE', 27, 33]]). Binary Indicators: Columns ending in *_bin* (e.g., *hedging_bin*) provide a 0 or 1 flag for the presence of those features. Metadata: Includes the annotator name and the data split (e.g., "train").

2. Annotation Schema (annotation_schema_forLLM.md) This Markdown document outlines the specific rules for identifying the five annotation tags used in the training set: *all_caps*: Matches words that are fully capitalized, excluding acronyms or single letters. *exclamation_marks*: Annotates any number of exclamation marks. *hedging*: Uses a specific lexicon to identify words expressing uncertainty; matches should be on the root of the word (e.g., presume). *adjectives*: Matches words with specific suffixes (-ful, -less, -ment, -ness, -ing, -ible) only if they are used as adjectives in context. *unk*: Specifically used for annotating profanity.

3. Lexicon of Hedging Words (hedge_words.txt) A plain text file containing a categorized list of words to be used for the hedging annotation. Categories include:

- Epistemic Verbs: e.g., suggest, believe, appear, think, suppose.
- Epistemic Adverbs: e.g., presumably, possibly, arguably, seemingly.
- Epistemic Adjectives: e.g., likely, probable, unsure, unclear.
- Epistemic Modal Verbs: e.g., may, might, could, would, should.
- Approximations: e.g., usually, approximately, generally.
- I am ready to assist with any tasks related to these datasets or the annotation process.

Prompt: Please give me 250 new annotated examples to add to the attached csv file. Let's do this in batches, starting with 60 first so I can check before we proceed.

- each example should have a new id, not duplicated from previous examples. please make the ids 6 digits of integers starting with "100xxx"
- each example should have a misinformation label that states if the example is misinformation or not, 0 for not, 1 for misinformation
- each example should have an opinion label that states if the example is opinion or not, 0 for not, 1 for opinion
- the text in each example should be no longer than 560 characters.
- the existing text examples are from twitter posts and news headlines, please follow those examples when creating new ones.
- each example should include annotations, in the same format as the existing annotations in the training set.
- please make the number of examples labeled misinformation and opinion balanced. There are four possible combinations of labels: 0/0, 0/1, 1/0, 1/1. Please give me an almost equal number of each combination.
- Generate examples that are similar in style and ambiguity as the below, with the correct label for each. Here are misinformation examples that our model got wrong:
  - 'Fear and panic in the air I want to be free from desolation and despair!'
  - 'The proportion of patients spending over 4 hours in A&E has substantially increased in recent years. Are waiting lists, ambulance data a'
  - 'Kids are inundated with images and information online and in media and have no way to deconstruct. - Kerri Sackville #TMS7'
  - '@afterShock_DeLo scuf ps live and the game... cya'

- 'The U.S. bombed Hiroshima 70 years ago today. A look at how war has changed since'
- 'Investigators say a fatal Virgin Galactic spaceship crash last year was caused by structural failure after the '
- 'I'm getting used to seeing a layer of ash on my car each morning - and I'm in AB. We've only seen a red sun this week. The smoke is as far as Sask.'
- 'Here's everything you need to know about the #InsurrectionAct'
- 'The #Berthoud Fall Classic Ride on 10/5 has been cancelled due to the floods destroying many roads on the route. The Oktoberfest will go on!'
- 'He lives in Palm Beach. Someone should check to see if he's got his hurricane shutters closed. And he ought to ride out Irma by his pool.'
- '@JoelOsteen refused to shelter hurricane victims in his "church", and NOW wants $$ for them. They will not see a penny. Such a hypocrite'
- 'Please check out our previous in-depth look at Abedin below:HILLARY S RIGHT-HAND WOMAN: HUMA ABEDIN S Job At A Radical Muslim Journal Calls Into Question Security Clearance For State DepartmentHILLARY S #1 AIDE HUMA ABEDIN: WHAT YOU NEED TO KNOW!'

Output: To get started on your request for 250 annotated examples, here is the first batch of 60. These examples follow the 6-digit ID format starting at 100001, maintain the 560-character limit, and utilize the specific annotation logic for *all_caps*, *exclamation_marks*, *hedging* (based on your lexicon), and *adjectives* (based on the specified suffixes). I have balanced the four label combinations (15 examples each).
(Produced examples not shown given length).

Prompt: We need more variety - look at the texts in the training_bs_al.csv and the examples I gave you in my last prompt. Some are very casual language with slang, profanity and abbreviations. Some are more formal and sound like reporting style.

Output: That makes perfect sense. I will increase the linguistic variety in this batch, incorporating more casual Twitter-style language (slang, abbreviations like "ppl," "bc," "tbh," and profanity for the *unk* category) alongside formal reporting and ambiguous news snippets. I have also ensured that the "wrongly classified" styles you shared—such as emotional outbursts, truncated news headlines, and highly specific local observations—are reflected here.
(Produced examples not shown given length).